ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** magnetic	Magnetic materials include iron, nickel and cobalt.
2	**(a)** small magnet	It is a tiny magnet.
3	**(a)** to keep them from losing magnetism	Keepers preserve magnetism.
4	**(d)** iron nail	Iron is magnetic.
5	**(d)** north-south	It aligns with Earth's field.
6	**(a)** mobile phones and credit cards	They may damage magnetic data.
7	**(d)** glass marble	Glass is not attracted.
8	**(c)** horseshoe magnet	U shaped.
9	**(a)** sailors	They find directions at sea.
10	**(d)** poles	Poles are at the ends.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Magnetic: iron, nickel, cobalt (1); Non-magnetic: wood, plastic, glass (1).	1+1
12	It is influenced by the Earth's magnetism (1); The Earth behaves like a large magnet (1).	1+1
13	They keep the magnetism from fading (1); They complete the magnetic pairs (1).	1+1
14	The ends where attraction is strongest (1); Called north and south poles (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Move a magnet over the mixture (1); Iron nails stick to the magnet (1); Wooden pieces remain (1).	1+1+1
16	Keep in pairs with keepers (1); Avoid heating and hammering (1); Keep away from electronic gadgets (1).	1+1+1
17	Roll the magnet in iron filings (1); Filings cling mostly at the ends (1); The ends are the poles (1).	1+1+1
	OR: Bring the magnet near the compass (1); Note which end attracts the north of the needle (1); That end is the south pole of the magnet (1).	1+1+1
18	Magnetic: attracted, e.g. iron (1); Non-magnetic: not attracted, e.g. plastic (1); Test with a magnet (1).	1+1+1
	OR: The nearby magnet attracts the needle (1); Its force is stronger than Earth's (1); Needle is deflected (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Collect various objects (1); Bring a magnet near each (1); Magnetic: iron nail, steel pin (1); Non-magnetic: wood, plastic, glass (1); Record in a table (1).	1+1+1+1+1
	OR: Magnetised needle on a pivot (1); Dial with directions (1); Hold it flat (1); Needle points north (1); Rotate the dial to match N (1).	1+1+1+1+1
20	Earth behaves like a bar magnet (1); A freely suspended magnet aligns north-south (1); Compass needle is a magnet (1); It settles in the north-south direction (1); Directions are found from this (1).	1+1+1+1+1
	OR: Spread iron filings on a sheet (1); Roll the bar magnet in them (1); Lift the magnet (1); Filings cling thickly near the ends (1); Very few cling to the middle (1).	1+1+1+1+1

Mistakes to avoid:

- Thinking everything metallic is magnetic; copper and aluminium are metals but **not** magnetic.
- Using attraction alone as the test for a magnet (iron is attracted too); the sure test is **repulsion**.
- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** bar	Bar magnets are common.
2	**(d)** it repels a known magnet	Only a magnet can repel another magnet.
3	**(d)** north and south	North-seeking and south-seeking poles.
4	**(c)** has two poles	Each piece is a magnet with N and S poles.
5	**(b)** even without touching it	Magnetic force attracts iron.
6	**(c)** north-south direction	It aligns with Earth's magnetic field.
7	**(c)** heated or hammered	Heating and hammering weaken them.
8	**(c)** the poles	Poles attract most iron filings.
9	**(a)** north pole	It is the north-seeking pole.
10	**(d)** rubbing with a magnet	Stroking with a magnet in one direction.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bar (1); Horseshoe or ring (1).	1+1
12	Each piece becomes a magnet (1); Each has a north and a south pole (1).	1+1
13	Like poles repel (1); Unlike poles attract (1).	1+1
14	Stroke it with a magnet in one direction (1); Repeat many times (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hang a bar magnet (1); Bring the same pole of another magnet near (1); It moves away, showing repulsion (1).	1+1+1
16	They attract (1); They may stick together (1); Unlike poles attract (1).	1+1+1
17	Refrigerator door (1); Pencil box closer (1); Speaker (1).	1+1+1
	OR: Iron, nickel, cobalt (1); Steel is used in tools (1); Magnets are used in speakers (1).	1+1+1
18	It is a small magnet (1); It turns freely (1); It settles in the north-south direction (1).	1+1+1
	OR: A force by a magnet (1); Acts without touching (1); Can attract or repel (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take a steel bar (1); Stroke with one pole of a magnet in one direction (1); Lift magnet at the end, repeat many times (1); Test using iron pins (1); The bar picks up pins (1).	1+1+1+1+1
	OR: Natural magnet called lodestone (1); It attracted iron (1); It pointed north-south when hung (1); Used for navigation (1); Led to the compass (1).	1+1+1+1+1
20	Suspend one bar magnet (1); Bring N pole of another near N pole: repulsion (1); Bring S pole near N pole: attraction (1); Like repel, unlike attract (1); Repulsion confirms a magnet (1).	1+1+1+1+1
	OR: Iron from sand (1); Iron from waste (1); Steel from scrap (1); Magnet picks only magnetic items (1); Others are left behind (1).	1+1+1+1+1

Mistakes to avoid:

- Thinking everything metallic is magnetic; copper and aluminium are metals but **not** magnetic.
- Using attraction alone as the test for a magnet (iron is attracted too); the sure test is **repulsion**.
- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** the ends	They cluster at the poles.
2	**(c)** repel each other	North repels north.
3	**(d)** iron, nickel, cobalt	These are the main magnetic metals.
4	**(d)** lodestone	Lodestone is a natural magnet.
5	**(c)** attract each other	North attracts south.
6	**(a)** aluminium foil	Aluminium is not attracted.
7	**(d)** it can act from a distance	Non-contact force.
8	**(b)** magnetic compass	A compass has a magnetised needle.
9	**(d)** magnetic force acts through paper	Magnetic force acts across non-magnetic materials.
10	**(d)** steel paper clip	Steel contains iron.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Compass (1); Door closers or fridge doors (1).	1+1
12	Its needle always points north-south (1); We find other directions from this (1).	1+1
13	Heating it (1); Hammering or dropping it (1).	1+1
14	They may damage the phone (1); They disturb its compass or memory (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No landmarks to guide (1); Needle gives north-south (1); Helps in finding direction (1).	1+1+1
16	Magnet repels one end of another magnet (1); Iron nail is only attracted (1); Repulsion shows a magnet (1).	1+1+1
17	Bring a magnet near (1); If attracted, magnetic (1); Else not (1).	1+1+1
	OR: Take a bar magnet (1); Stroke needle from end to end in one direction (1); Test by picking up pins (1).	1+1+1
18	Place paper between magnet and pin (1); Pin is still attracted (1); Force passes through non-magnetic sheets (1).	1+1+1
	OR: Hang the magnet freely (1); The end that points north is north pole (1); The other end is south pole (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heating (1); Hammering (1); Dropping (1); Keep in pairs (1); Place keepers (1).	1+1+1+1+1
	OR: Each piece is a magnet (1); Each has N and S pole (1); Smaller pieces remain magnets (1); Poles cannot be isolated (1); Strength decreases (1).	1+1+1+1+1
20	Compass (1); Fridge door (1); Loudspeaker (1); Cranes lifting scrap iron (1); Magnetic toys (1).	1+1+1+1+1
	OR: Attracts magnetic materials (1); Has two poles (1); Like poles repel (1); Unlike poles attract (1); Points north-south when free (1).	1+1+1+1+1

Mistakes to avoid:

- Thinking everything metallic is magnetic; copper and aluminium are metals but **not** magnetic.
- Using attraction alone as the test for a magnet (iron is attracted too); the sure test is **repulsion**.
- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

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